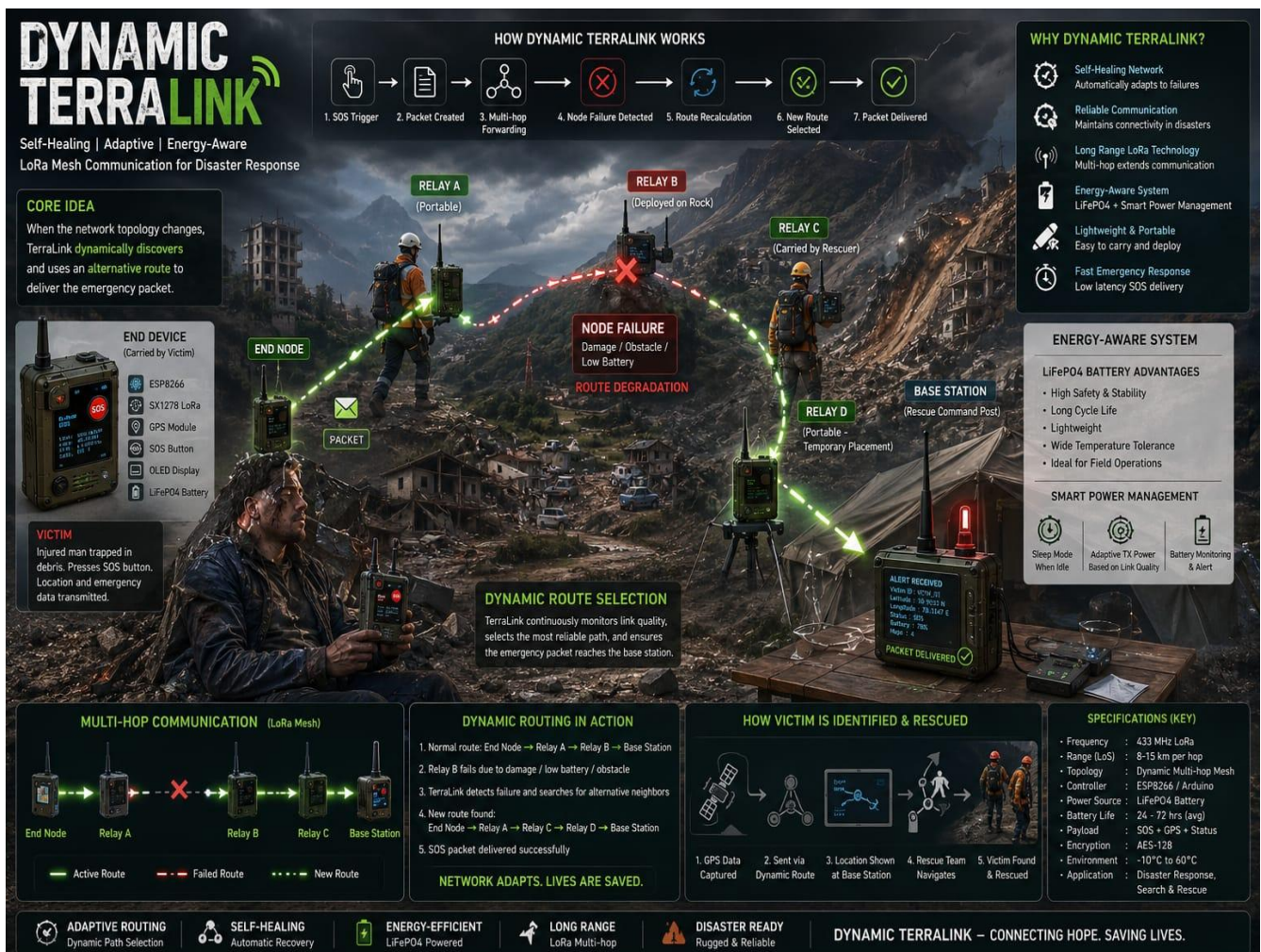


PROPOSED SOLUTION

- **Dynamic TerraLink** is a LoRa-based, multi-hop emergency communication system designed to maintain connectivity when conventional communication infrastructure is unavailable.
- TerraLink dynamically adapts its communication route when nodes or links become unavailable, enabling continued emergency communication without relying on cellular infrastructure.

Core Solution

- **SOS + GPS:** User triggers an SOS and sends their location.
- **LoRa Communication:** Provides long-range, low-power wireless communication.
- **Multi-Hop Relays:** Emergency packets travel through multiple relay nodes.
- **Dynamic Routing:** Nodes discover and select suitable communication paths.
- **Route Monitoring:** The network monitors node and link conditions.
- **Self-Healing:** If a relay or link fails, an alternative route is discovered and selected.
- **Emergency Priority:** SOS packets are given priority during network congestion.
- **Reliable Delivery:** The emergency message and location are delivered to the rescue/base station.



FUNCTIONAL REQUIREMENTS:

1. **SOS Generation** – Trigger SOS and create emergency packet.
2. **GPS Acquisition** – Capture and transmit location.
3. **LoRa Communication** – Send/receive emergency data.
4. **Multi-Hop Forwarding** – Forward packets through relays.
5. **Neighbour Discovery** – Identify available relay nodes.
6. **Dynamic Route Discovery** – Find alternative routes.
7. **Route Selection** – Select a suitable/reliable path.
8. **Failure Detection** – Detect failed/degraded links or nodes.
9. **Route Recovery** – Switch to an alternative route.
10. **Emergency Prioritization** – Prioritize SOS packets.
11. **Duplicate Filtering** – Prevent repeated forwarding.
12. **TTL / Hop Control** – Limit packet propagation.
13. **Node Monitoring** – Monitor battery, link, and availability.
14. **Emergency Delivery** – Deliver SOS + GPS to the base station.

